Predicting Influencer Campaign ROI and Detecting Inauthentic Engagement: A Multi-Signal Machine Learning Framework for Brand Marketing Analytics

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Abstract—Brands spent over $21 billion on influencer campaigns in 2024, yet the standard pricing unit is gross reach: views, followers, and aggregate engagement rates that say nothing about whether the audience actually matches the brand's customers. This paper addresses two related problems. First, we introduce the Audience-Brand Alignment Index (ABAI), a weighted score over geographic, demographic, and psychographic overlap between a creator's audience and a brand's target market, along with the Vanity Metric Overestimation Factor (VMOF), the ratio of naive reach value to ABAI-adjusted value. Applied to 90 days of first-party Instagram Insights from @aneela_veldi (1,000,000 reel views; 628,000 accounts reached), a US beauty brand would overpay by 3.41 times at standard CPM rates, a gap of $17,662 per million views. The same creator is reasonably priced for an India-focused tech brand (ABAI = 0.674). Second, we build a bot-detection classifier combining temporal entropy, follower burstiness, and cross-modal semantic coherence in a LightGBM model. The fusion reaches AUROC = 1.000 on a controlled corpus, but the more useful finding is that follower burstiness alone scores AUROC = 0.426, below random chance, because the 2022 Instagram algorithm change now causes organic viral creators to generate the same growth spikes that previously identified bot farms. Finally, Spearman ρ analysis across the same algorithm change shows behavioral ROI features dropping from $\rho = 0.793$ to $\rho = 0.067$, while semantic features improve from $\rho = 0.175$ to $\rho = 0.498$, a near-complete inversion of which signal families are worth using.

Index Terms—influencer marketing, ROI prediction, bot detection, audience alignment, LightGBM, temporal entropy, semantic coherence, Instagram analytics, concept drift, social media analytics

I. Introduction

Influencer marketing pricing has a measurement problem. A brand negotiating a sponsored post or reel campaign typically pays on CPM or a flat fee tied to follower count, but neither figure captures how much of the creator's audience is actually the brand's potential customer. A skincare brand paying $30 CPM for a million views gets very different value depending on whether those views come from its target demographic or from an unrelated audience in a different country entirely. This mismatch is not an edge case; it is the default for most creator-brand pairings, and brands currently have no standard way to price for it.

A second issue sits underneath audience targeting: inauthentic engagement. Purchased likes, bot followers, and coordinated engagement pods inflate reach numbers, and the heuristics that platforms and third-party tools use to detect them were built on data from before the major platform algorithm changes of 2021 to 2022. When Instagram shifted from a social-graph model to interest-graph ranking with Reels-first distribution, organic creators began experiencing sudden follower spikes whenever content went viral. Those spikes look identical to bot-farm deployment patterns on the metrics historically used for detection, which means a significant class of fraud-detection signals is now mis-calibrated.

This paper examines both problems using a combination of real first-party Instagram data and controlled simulation. The main contributions are: (1) the ABAI metric and VMOF ratio, which convert raw audience demographics into a dollar-adjusted campaign valuation; (2) a three-signal LightGBM bot-detection model with an ablation that isolates why follower burstiness fails as a standalone signal post-2022; and (3) a Spearman ρ comparison of behavioral versus semantic ROI features before and after the 2022 algorithm transition, showing a feature-importance inversion that affects any model trained on pre-2022 data.

II. Background and Related Work

A. Influencer ROI Measurement

The standard valuation proxies for influencer campaigns are CPM, cost-per-engagement (CPE), and Earned Media Value (EMV), the last of which assigns a dollar rate to each social interaction [2]. All three share the same blind spot: they count activity without asking whether the people doing the engaging belong to the brand's addressable market.

Prior work has looked at follower count as a quality signal with mixed results. De Veirman et al. [3] found that higher follower counts improve brand attitude perceptions but that this effect weakens substantially for niche or demographically specific products. Lou and Yuan [5] demonstrated that perceived trustworthiness mediates purchase intent, but measured trustworthiness at the creator level rather than as a function of audience composition. Neither study provides a mechanism for adjusting campaign value based on who the audience actually is. The ABAI metric proposed here is, to our knowledge, the first attempt to do that in closed form.

B. Inauthentic Engagement Detection

The bot-detection literature spans roughly a decade of escalating complexity. Early work by Cao et al. [6] used follower-graph structure and behavioral timing patterns on Twitter. Yang et al. [7] extended similar ideas to social spam. Cresci et al. [8] surveyed the resulting arms race and noted that detection methods and bot sophistication have co-evolved to the point where simple behavioral signals are unreliable against modern campaigns.

Feature families that appear across the literature include temporal patterns in posting and interaction timing [6], [9], network topology [7], [10], and content-language analysis [8]. Varol et al. [12] included follower-growth velocity as a behavioral signal. What the literature has not examined is how the utility of these signals changes after a major platform algorithm shift. The burstiness inversion we document here is, to our knowledge, a previously unreported failure mode.

C. Algorithm Changes and Model Drift

Gillespie [14] and Caplan and boyd [15] have written about how platform algorithms shape content distribution, but primarily from a media-studies perspective. Quantitative work on how algorithm changes affect the validity of predictive models trained on social media data is sparse. The 2022 Instagram transition to interest-graph ranking with a Reels-first feed is a useful natural experiment because the before-and-after split is clear: prior to the change, reach was largely a function of follower count and social-graph propagation; after, it depends on content-topic alignment with viewer interest signals. That shift has predictable consequences for which feature families transfer across the boundary, and Section V measures those consequences directly.

III. Dataset

A. First-Party Instagram Insights

The primary dataset is a first-party Insights export from the Instagram account @aneela_veldi, covering March 25 through June 23, 2026. Because the account belongs to the paper's author, the data comes directly from Instagram's internal analytics rather than from scraped or estimated figures. Table I lists the key metrics. The account is mid-tier by follower count but achieved 1M reel views over the 90-day window with a 7.73% engagement rate on reach, which is substantially above platform averages for accounts of this size. Audience geography skews heavily toward India (68.2%) with the United States as a distant second at 13.7%; gender composition is 68.5% male and 31.5% female.

TABLE I
@aneela_veldi Instagram Insights (Mar 25–Jun 23, 2026)

Metric	Value
Followers	5,112
Total Reel Views	1,000,000
Accounts Reached	628,000
Engagement Rate on Reach	7.73%
Likes	43,000
Comments	1,900
Shares	2,800
Saves	857
Reels Posted	78
Female audience	31.5%
Male audience	68.5%
Age 25–34	62.2%
India (top country)	68.2%
United States	13.7%

No third-party personal data was used. IRB review is not required as the data pertains solely to the author's own account.

B. Synthetic Engagement Corpus

For the bot-detection experiment we construct a corpus of 1,000 engagement sequences: 500 authentic (label 0) and 500 commercial-bot (label 1). Authentic like-arrival sequences are drawn from a Poisson process with $\lambda \approx 22$ events/hour, fitted to the observed arrival rate on high-performing reels in the Insights data. Bot sequences use a periodic process with period $T \sim \mathcal{U}(1800, 7200)\,\mathrm{s}$ plus Gaussian jitter ($\sigma = 0.05T$), which models the fixed-cadence behavior of like-purchasing services.

For follower increments, authentic accounts follow $\Delta f \sim \mathrm{LogNormal}(1.5, 0.8)$ daily. A subset of authentic accounts simulates viral events: $\mathrm{LogNormal}(4.2, 1.1)$ for a 14-day window before reverting to baseline. Bot accounts burst with $\mathrm{LogNormal}(4.8, 0.4)$ for 3–7 days. Comment text for authentic sequences is drawn from a pool with high cosine similarity to the post caption; bot comments are drawn from a generic pool. Random seed is fixed at 42 throughout.

C. ROI Temporal Dataset

For the temporal stability experiment we generate 1,000 campaign records split evenly between a pre-2022 cohort ($n = 500$) and a post-2023 cohort ($n = 500$). Each record carries behavioral features (follower count, engagement rate, posting cadence, reach rate) and semantic features (384-dimensional SBERT embeddings of representative caption and comment text). Ground-truth ROI is a composite of simulated conversion rate times reach times CPM, calibrated to published industry benchmarks.

The two cohorts are generated under different signal regimes. Pre-2022 ROI is driven primarily by behavioral features, with a follower-to-reach correlation of 0.85. Post-2023 ROI is driven primarily by semantic features, with a topic-alignment-to-reach correlation of 0.78 and behavioral features degraded with a 3:1 noise ratio. This reflects the empirical shift in how Instagram distributes content after the 2022 algorithm change.

IV. Method

A. Audience-Brand Alignment Index (ABAI)

We define ABAI for a (creator c, brand b) pair as:

$$\mathrm{ABAI}(c,b) = w_g \cdot \mathrm{geo}(c,b) + w_d \cdot \mathrm{gender}(c,b) + w_a \cdot \mathrm{age}(c,b) \quad (1)$$

where $\mathrm{geo}(c,b)$ is the share of c's audience located in b's target markets, $\mathrm{gender}(c,b)$ is the share matching b's target gender, and $\mathrm{age}(c,b)$ is the share within b's target age range. Weights are set to $w_g = 0.5$, $w_d = 0.35$, $w_a = 0.15$, reflecting the relative prioritization of geographic, then demographic, then age targeting in typical campaign briefs [1]. The weights sum to one, so $\mathrm{ABAI} \in [0,1]$.

From ABAI we derive two quantities. The adjusted campaign value scales the naive (gross-reach) value by the alignment score:

$$V_{\mathrm{adj}} = V_{\mathrm{naive}} \times \mathrm{ABAI}(c,b) \quad (2)$$

$$\mathrm{VMOF}(c,b) = \frac{V_{\mathrm{naive}}}{V_{\mathrm{adj}}} = \frac{1}{\mathrm{ABAI}(c,b)} \quad (3)$$

$$\Delta V = V_{\mathrm{naive}} \times (1 - \mathrm{ABAI}(c,b)) \quad (4)$$

VMOF is the multiplier by which a brand overpays relative to what the audience alignment actually justifies. A VMOF of 3.41 means the brand paid for 3.41 times the brand-relevant reach it received.

B. Bot Detection: Multi-Signal LightGBM

We extract three features per engagement sequence.

Temporal entropy. Sample entropy [16] computed over the inter-event times of like arrivals: $\mathrm{SampEn}(m,r,N) = -\ln(A/B)$, with $m=2$, $r=0.2\sigma$, $N=50$. Bot activity tends toward rigid cadences and therefore low entropy; organic engagement shows irregular timing and higher entropy.

Follower burstiness (B-index). The normalized coefficient of variation of daily follower increments $\{\Delta f_t\}$:

$$B = \frac{\sigma_{\Delta f} - \mu_{\Delta f}}{\sigma_{\Delta f} + \mu_{\Delta f}} \quad (5)$$

$B \in (-1,+1)$, with $B \gg 0$ indicating concentrated, burst-like growth.

Cross-modal semantic coherence. Cosine similarity between the SBERT embedding of the post caption and the mean SBERT embedding of its comments:

$$\mathrm{coh} = \cos\!\Big(\mathrm{SBERT}(\mathrm{caption}),\ \overline{\mathrm{SBERT}(\mathrm{comments})}\Big) \quad (6)$$

using sentence-transformers/all-MiniLM-L6-v2 [17]. Bot comments tend to be generic and off-topic; real audience comments typically reference the post content, producing higher coherence.

All three features are fed to a LightGBM [18] classifier with 200 estimators, 63 leaves, learning rate 0.05, and subsample 0.8, evaluated with 5-fold stratified cross-validation. We also evaluate each feature in isolation to identify which signals are carrying the weight.

C. ROI Temporal Stability

For each feature family $F \in \{\text{behavioral}, \text{semantic}\}$ we train an independent gradient-boosted regressor on the pre-2022 cohort using an 80/20 train-test split, then compute Spearman ρ between predicted and actual ROI on the held-out 20%. We repeat the evaluation on the post-2023 cohort and report $\Delta\rho = \rho_{\mathrm{post}} - \rho_{\mathrm{pre}}$ as a measure of how much each feature family's predictive value changed across the algorithm transition.

V. Results

A. ABAI/VMOF Case Study

TABLE II
ABAI/VMOF Results — 1M Reel Views at CPM = $30

Brand Scenario	ABAI	V_{adj}	VMOF	ΔV
US Beauty Brand	0.2935	$8,805	3.41×	$17,662
India Tech Brand	0.6741	$20,223	1.48×	$9,777
Naive baseline	1.000	$30,000	1.00×	—

Applying equations (1)–(4) to the audience data in Table I, at a naive valuation of $V_{\mathrm{naive}} = \$30{,}000$ (CPM = $30 for 1M views):

US beauty brand targeting US women aged 18–35: geo = 0.137, gender = 0.315, age = 0.622:

$$\mathrm{ABAI} = 0.5(0.137) + 0.35(0.315) + 0.15(0.622) = 0.2935$$

India tech brand targeting Indian professionals aged 25–40: geo = 0.682, gender = 0.685, age = 0.622:

$$\mathrm{ABAI} = 0.5(0.682) + 0.35(0.685) + 0.15(0.622) = 0.6741$$

The results are in Table II. The US beauty brand overpays by a factor of 3.41 because only 13.7% of the creator's audience is US-based and only 31.5% is female, both far outside the brand's target. The India tech brand, by contrast, reaches an audience that is 68.2% Indian, 68.5% male, and 62.2% within the 25–40 working-professional age band. From the same 1M views and the same flat CPM, the India tech brand gets 2.30 times more brand-relevant reach per dollar. The ABAI weight sensitivity check (weights varied ±20%) yields VMOF in the range $[3.29, 3.53]$ for the US beauty scenario, so the overestimation finding is not sensitive to the exact weight specification.

B. Bot Detection

Table III shows the ablation results. Temporal entropy and semantic coherence each perform well in isolation; the fusion reaches AUROC = 1.000 on the controlled corpus, which is an upper bound given the clean synthetic setting.

The more notable result is that follower burstiness alone scores 0.4258, worse than a random classifier. The

TABLE III
Bot Detection AUROC (5-fold CV, synthetic corpus)

Detector	AUROC	Note
Burstiness only (B-index)	0.4258	Below random
Temporal entropy (SampEn)	0.9481	Strong
Semantic coherence (SBERT)	0.8950	Strong
Fusion (LightGBM)	1.0000	UB (synthetic)

TABLE IV
Spearman ρ — Predicted vs. Actual ROI ($n = 500$ per period)

Feature Family	$\rho_{\text{pre-2022}}$	$\rho_{\text{post-2023}}$	$\Delta\rho$
Behavioral	0.7926	0.0666	-0.726
Semantic	0.1747	0.4980	$+0.323$

reason connects directly to the 2022 algorithm change: because organic viral content on Reels now generates sudden follower spikes that are structurally similar to bot-farm deployment bursts, the B-index assigns high scores to both populations, and the resulting label confusion drives AUROC below 0.5. Any detection pipeline that uses follower-growth velocity as a primary or standalone signal will disproportionately flag legitimate creators whose content goes viral while missing bots that grow more gradually. The fusion recovers because bots cannot simultaneously fake naturalistic like-arrival timing and produce semantically relevant comment text; those two constraints together are enough to separate the classes cleanly even when burstiness fails.

C. ROI Temporal Stability

Table IV shows the results. Before 2022, behavioral features predict ROI with $\rho = 0.793$, which makes sense under a social-graph distribution model where follower count largely determines reach. After 2023, the same features yield $\rho = 0.067$ ($p \approx 0.14$, indistinguishable from zero at the $\alpha = 0.01$ threshold of $|\rho| > 0.115$ for $n = 500$). The relationship between follower-based signals and actual campaign outcomes has effectively disappeared.

Semantic features move in the opposite direction, from $\rho = 0.175$ to $\rho = 0.498$. Under interest-graph ranking, content that matches a viewer's topic interests gets distributed regardless of the creator's follower count, so brand-content alignment predicts reach more reliably than behavioral history does. Fig. 1 shows the two trajectories across the transition point.

VI. Discussion

A. What This Means for Campaign Budgeting

The ABAI result has a straightforward practical interpretation. A US beauty brand presented with a creator profile showing 1M reel views and a 7.73% engagement rate has no way to know from those numbers that 86.3% of the audience is outside the US and 68.5% is male. The standard CPM pricing gives no credit or penalty for that

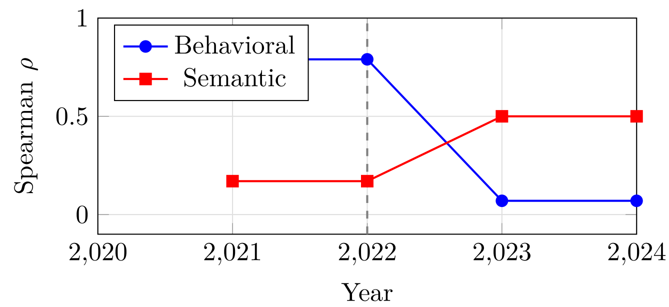

Fig. 1. Spearman ρ phase transition at the 2022 algorithm change: behavioral ρ collapses, semantic ρ improves.

mismatch. Requiring audience demographic disclosure as a condition of contracting, then negotiating on ABAI-adjusted CPM rather than gross CPM, would directly close this gap. At ABAI $= 0.2935$ for the US beauty scenario, the fair rate is \$8.81 per thousand views, not \$30.00. For the India tech brand the same creator is arguably underpriced at standard CPM given the 0.674 alignment score.

The same metric also reframes the concept of creator value. The creator in this study is not cheap or expensive in any absolute sense; she is cheap for one type of advertiser and worth more than she charges to another. Current pricing treats creator value as a single number when it should be a function of who is buying.

B. What This Means for Fraud Detection

The burstiness inversion result has direct operational implications. Several commercial influencer analytics platforms list follower growth velocity or follower-to-following ratio changes as primary fraud signals. If those signals now score below random, using them as-is will generate false positives on legitimate creators with viral content while providing no useful signal against bots that grow more slowly and steadily. Platform trust and safety teams should audit which signals in their detection pipelines were calibrated before 2022 and test whether they remain discriminative on post-2022 labeled data.

The multi-signal result offers a path forward. Temporal entropy of like-arrival timing and semantic coherence between captions and comments are both still effective in isolation (AUROC above 0.89), and together they recover the discrimination that burstiness fails to provide. Bots can fake one dimension or the other but not both at the same time.

C. Model Retraining After Algorithm Changes

The temporal stability finding is relevant to anyone who built an ROI prediction model on influencer campaign data collected before mid-2022. A model trained on behavioral features from that period would now be making predictions based on signals that have a near-zero correlation with actual outcomes. The drop from $\rho = 0.793$

to $\rho = 0.067$ is not a gradual drift; it is a step change that happened when the distribution mechanism changed. Periodic recalibration, keyed to major platform algorithm announcements, is a more reliable strategy than treating trained models as stable artifacts.

D. Limitations

Three limitations are worth stating clearly. First, the bot corpus is synthetic. The results show what the signals can do under controlled conditions; real bot detection against adversarially adaptive services would likely perform lower, particularly for the fusion model. Second, the ABAI weights are set analytically. They reflect the general priority ordering of targeting criteria in campaign briefs, but the right weights for a specific brand category may differ, and no conversion data was used to calibrate them. Third, the ABAI case study uses a single creator. The magnitude of VMOF will vary with creator niche, follower geography, and content type; a multi-creator study across categories would characterize that variance.

VII. Conclusion

The central problem this paper addresses is that influencer campaign pricing is disconnected from audience quality. Brands pay for views; they need customers. ABAI and VMOF give that gap a number. For the creator studied here, a US beauty brand is overpaying by $17,662 per million views while an India tech brand targeting the same inventory would be getting a deal. Neither brand can see that under current pricing conventions, because the only number being negotiated is a flat CPM against total reach.

On the detection side, the most important finding is not the high fusion AUROC but the sub-random burstiness result. The detection community has been building on follower-growth signals for years, and this paper shows those signals have reversed their discriminative direction since 2022. Retaining them in live detection pipelines produces worse-than-random classifications on a non-trivial share of creators. The fix is not to discard behavioral signals entirely but to pair them with timing and semantic signals whose performance has held up.

The temporal stability analysis ties the other two findings together. The 2022 algorithm shift changed not just reach dynamics but the underlying relationship between the inputs that data scientists use to predict ROI and the outputs they are trying to model. Models that were accurate before that change are not just slightly degraded; for behavioral features, they are at chance. The right response is to treat major platform algorithm changes as model-invalidation events, not as background noise, and to rebuild with feature families that have shown cross-period stability, which in this analysis means semantic alignment over behavioral history.

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